

MATH 332 Linear Algebra: Theory and Applications

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Test 1

Thursday, September 17, 2026

Instructions:

- i. This test has **FOUR questions**. Attempt all. The maximum number of points is **100**.
- ii. The time allowed is 75 minutes.
- iii. This test is closed book, but you may use **4** double-sided letter-size sheets of notes.
- iv. No calculators or other electronic devices are allowed. Phones must be placed in your bags under your desks or at the front of the room. Hands must be on top of your desks.
- v. You are expected to simplify any expressions that can be simplified as whole numbers, e.g., $36/15 = 12/5$, but you may leave other expressions as is, e.g., $\sqrt{47}$ or $18/29$.
- vi. Show all your work to justify your answers. Answers without adequate justification will not receive credit.
- vii. Print your name clearly on each answer sheet.
- viii. Off-site students may contact the instructor as directed by your syllabus.

Q	Max	Score
1	16	
2	33	
3	30	
4	21	
Tot	100	

ChatGPT/Codex assisted in the construction of this exam.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

Signature

Date

Printed Name

1. (16 points) Determine whether each statement is true or false. For each true statement, name the rule that makes it true or give a confirming example. For each false statement, explain the error or give a counterexample. A brief justification is sufficient.
 - a. (4 points) A valid row operation is $R_1 \leftarrow 2R_1$.

Answer: True. Multiplying a row by any nonzero scalar is a valid row operation.

- b. (4 points) A valid row operation is to replace every entry in R_1 by its square.

Answer: False. Replacing every entry in a row by its square is not one of the elementary row operations and would not, in general, preserve the solution set of a system.

- c. (4 points) In general, $\mathbf{AB} = \mathbf{BA}$ whenever both products are defined.

Answer: False. Matrix multiplication is not commutative in general. For example, with

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

we have

$$\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \mathbf{BA}.$$

- d. (4 points) A system of two linear equations in one unknown can have a unique solution.

Answer: True. For example, the system

$$x = 1, \quad 2x = 2$$

has the unique solution $x = 1$.

2. (33 points) The augmented matrix for the system $\mathbf{Ax} = \mathbf{b}$ is

$$[\mathbf{A} \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ 1 & 2 & 3 & 9 \end{array} \right].$$

Starting with this matrix, use forward elimination and back substitution to solve the system. Show your row operations.

Answer: The zero first pivot requires a row swap:

$$\begin{aligned} \left[\begin{array}{ccc|c} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ 1 & 2 & 3 & 9 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 2 & 3 & 9 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]. \end{aligned}$$

Back substitution gives

$$\begin{aligned} z &= 3, \\ y + z &= 2 \implies y = -1, \\ x + y + z &= 4 \implies x = 2. \end{aligned}$$

Thus the unique solution is

$$\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

Suggested grading: row swap and forward elimination, 20 points; back substitution and solution, 13 points.

3. (30 points) For the matrix $\mathbf{A}$ in the system $\mathbf{Ax} = \mathbf{b}$ from Problem 2, complete the following.
- a. (26 points) Find $\mathbf{A}^{-1}$. Show your row operations.

Answer: Row reduce the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$:

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & -1 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right]. \end{aligned}$$

Finish the Gauss–Jordan elimination with

$$R_2 \leftarrow R_2 - R_3, \quad R_1 \leftarrow R_1 - R_3, \quad R_1 \leftarrow R_1 - R_2.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right] = [\mathbf{I} \mid \mathbf{A}^{-1}], \quad \mathbf{A}^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}.$$

Suggested grading: correct $[\mathbf{A} \mid \mathbf{I}]$ setup, 4 points; forward elimination, 10 points; reduction to $[\mathbf{I} \mid \mathbf{A}^{-1}]$ and final inverse, 12 points.

- b. (4 points) Use your inverse to compute $\mathbf{A}^{-1}\mathbf{b}$, and verify that it agrees with your solution to Problem 2.

Answer:

$$\mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} -2 + 4 \\ 4 + 4 - 9 \\ -2 - 4 + 9 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

This agrees with the solution to Problem 2.

4. (21 points) Consider the three matrices

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Identify which matrix represents each of the following transformations. Explain each answer.

- a. (7 points) A permutation.

Answer: A_3 is a permutation matrix: it is obtained by exchanging the two rows of I_2 , and it exchanges the two coordinates.

- b. (7 points) A projection.

Answer: A_1 is the projection onto the x -axis, since

$$A_1 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}.$$

Its output lies on the x -axis, it fixes every vector on that axis, and $A_1^2 = A_1$.

- c. (7 points) A rotation.

Answer: A_2 is rotation counterclockwise through 90° , since

$$A_2 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}.$$

Test 1 Scores (out of 100)

Number of Students: 20, Minimum: 46, Maximum: 100, Mean: 87.1, Median: 96

Standard Deviation: 17.7, Quartiles (Q1, Q3): (76, 100)

10		000000000
9		2366
8		06
7		2
6		46
5		1
4		6